Highlights

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- Enriched 36-dim atom features and 12-dim bond features from RDKit consistently improve GNN molecular property prediction
- Directed message passing with bond features outperforms undirected GNNs on 3 of 4 MoleculeNet benchmarks
- DMPNN with enriched features achieves best GNN AUROC on BBBP (0.831), BACE (0.755), and HIV (0.678)
- Ablation study shows enriched atom features and bond features contribute independently and synergistically
- Systematic comparison of 7 GNN baselines with scaffold splitting and multiple random seeds

Enriched Molecular Features with Directed Message Passing for Property Prediction on MoleculeNet[⋆]

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ABSTRACT

Molecular property prediction is a fundamental task in drug discovery and computational chemistry. Graph neural networks (GNNs) have become the dominant approach, yet standard architectures typically ignore bond features during message passing and use undirected communication schemes that can lead to information redundancy. In this paper, we systematically investigate whether combining enriched molecular features with directed message passing can improve GNN performance on MoleculeNet benchmarks. We augment the Directed Message Passing Neural Network (DMPNN) with 36-dimensional RDKit-computed atom features and 12-dimensional bond features that encode detailed chemical information including atom type, hybridization, formal charge, chirality, bond stereochemistry, and ring membership. We evaluate the enriched-feature DMPNN against six GNN baselines (GCN, GAT, GIN, GraphSAGE, MPNN, DMPNN with standard features) and a Random Forest baseline on Morgan fingerprints, across four MoleculeNet benchmarks—BBBP, BACE, HIV, and Tox21—using scaffold splitting. Our results demonstrate that enriched features combined with directed message passing consistently outperform standard GNN approaches, achieving the best GNN performance on three of four benchmarks (BBBP: 0.831 AUROC; BACE: 0.755 AUROC; HIV: 0.678 AUROC) with notably low variance on the HIV dataset (standard deviation of 0.006). An ablation study confirms that both enriched atom features and directed bond-aware message passing contribute independently to performance gains, with the combination yielding the largest improvements. These findings underscore the importance of feature engineering and message passing design in molecular graph learning, and provide practical guidance for practitioners building molecular property prediction systems.

1. Introduction

Molecular property prediction is a cornerstone of modern drug discovery and computational chemistry, enabling researchers to estimate the biological activity, toxicity, and pharmacokinetic properties of candidate compounds from their structural representations alone (Wu, Ramsundar, Feinberg, Gomes, Geniesse, Pappu, Leswing and Pande, 2018). Accurate *in silico* prediction of properties such as blood–brain barrier penetration (BBBP), β-secretase 1 (BACE-1) binding affinity, and human immunodeficiency virus (HIV) inhibition can dramatically accelerate the drug development pipeline by prioritizing compounds for expensive and time-consuming wet-laboratory experiments (Yang, Swanson, Jin, Coley, Eiden, Gao, Guzman-Perez, Hopper, Kelley, Mathea et al., 2019). Traditional approaches relied on hand-crafted molecular descriptors and fixed-size fingerprints, such as Morgan/ECFP fingerprints (Rogers and Hahn, 2010), which encode the local chemical environment of each atom into a fixed-length bit vector. While effective, these fingerprint-based methods require extensive domain expertise for feature selection and often fail to capture complex non-linear structure–property relationships (Weininger, 1988).

The advent of deep learning has transformed molecular property prediction. In particular, graph neural networks (GNNs) have emerged as the dominant paradigm (Wu, Pan, Chen, Long, Zhang and Yu, 2021; Zhang, Chen, Li, Pan, Chen, Li and Chen, 2020). GNNs naturally represent molecules as graphs, where atoms

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correspond to nodes and chemical bonds correspond to edges. This graph-based representation preserves the topological structure of molecules and allows GNNs to learn hierarchical features through iterative message passing between neighboring atoms (Gilmer, Schoenholz, Riley, Vinyals and Dahl, 2017). Prominent GNN architectures—including Graph Convolutional Networks (GCNs) (Kipf and Welling, 2017), Graph Attention Networks (GATs) (Veličković, Cucurull, Casanova, Romero, Liò and Bengio, 2018), Graph Isomorphism Networks (GINs) (Xu, Hu, Leskovec and Jegelka, 2019), and GraphSAGE (Hamilton, Ying and Leskovec, 2017)—have demonstrated strong performance on various molecular benchmarks.

Despite this progress, two significant limitations persist in existing GNN architectures for molecular property prediction, both of which stem from incomplete utilization of the chemical information available in molecular graphs.

First, most standard GNN architectures treat edges as unweighted adjacency indicators, discarding the rich chemical information encoded in bond features. Chemical bonds carry critical information beyond simple connectivity: bond type (single, double, triple, aromatic) determines the strength and nature of atomic interactions; stereo configuration (E/Z, cis/trans) influences three-dimensional molecular shape; conjugation affects electronic delocalization; and ring membership reflects structural rigidity. The Message Passing Neural Network (MPNN) framework of Gilmer et al. (2017) permits edge features in principle, but many practical instantiations—including GCN, GAT, GIN, and GraphSAGE—do not exploit them. This omission forces the network to infer bond properties implicitly from atom features and graph topology alone, which is fundamentally limited.

Second, standard GNN message passing is undirected in the sense that messages flow simultaneously along both directions of each edge. As demonstrated by Yang et al. (2019), this can lead to informational redundancy: a message sent from atom i to atom j at layer t may be immediately returned to atom i at layer $t + 1$, effectively wasting a message passing step. The Directed Message Passing Neural Network (DMPNN) addresses this by associating messages with directed edges rather than nodes, ensuring that each message carries genuinely new information from the sender to the receiver.

In this paper, we conduct a systematic investigation of whether combining enriched molecular features with directed message passing can improve GNN performance on standardized molecular property prediction benchmarks. Our approach augments the DMPNN architecture of Yang et al. (2019) with two key enhancements to the input representation:

1. **Enriched atom features**: We replace simplistic one-hot atom encodings (typically 9–14 dimensions) with 36-dimensional RDKit-computed feature vectors encompassing element type, degree, formal charge, hybridization, aromaticity, total hydrogen count, chirality, and ring membership. These features provide the GNN with substantially more chemical context at the input level.
2. **Enriched bond features**: We encode each chemical bond as a 12-dimensional feature vector covering bond type, stereo configuration, conjugation, and ring membership. These features are injected into the directed message function, ensuring that bond chemistry directly modulates information flow during message passing.

We evaluate our enriched-feature DMPNN on four MoleculeNet benchmarks (Wu et al., 2018)—BBBP, BACE, HIV, and Tox21—using scaffold splitting to simulate realistic drug-discovery evaluation. We compare against six GNN baselines (GCN, GAT, GIN, GraphSAGE, MPNN, and DMPNN with standard features) and a Random Forest classifier on Morgan fingerprints. All GNN baselines use identical hidden dimensions and training procedures to ensure fair comparison.

Our key findings are as follows:

- Enriched-feature DMPNN achieves the best GNN performance on three of four benchmarks: BBBP (AUROC 0.831), BACE (AUROC 0.755, +13 points over the second-best GNN), and HIV (AUROC 0.678, with the lowest standard deviation of 0.006 among all methods).

- The directed message passing mechanism with bond features consistently outperforms undirected GNN architectures that ignore edge information.

- Both enriched atom features and enriched bond features contribute independently to performance, with the largest gains observed when both are combined.

- Random Forest on Morgan fingerprints remains a strong baseline on binary classification tasks, highlighting the continued value of established cheminformatics methods.

The remainder of this paper is organized as follows. Section 2 reviews related work on graph neural networks for molecular property prediction. Section 3 describes the DMPNN architecture and our enriched feature representations. Section 4 presents the experimental setup, results, and ablation studies. Section 5 discusses the implications of our findings and the reasons behind the observed performance patterns. Finally, Section 6 concludes with practical recommendations and future directions.

2. Related Work

2.1. Graph Neural Networks for Molecular Property Prediction

Graph neural networks have become the standard approach for molecular property prediction (Wu et al., 2021). The foundational work by Duvenaud, Maclaurin, Iparraguirre, Bombarell, Hirzel, Aspuru-Guzik and Adams (2015) introduced neural graph fingerprints, which generalize traditional molecular fingerprints by learning atom-level features through differentiable message passing. Gilmer et al. (2017) proposed the Message Passing Neural Network (MPNN) framework, which unifies various GNN architectures under a common formulation: at each layer, messages are computed from neighboring node features, aggregated, and used to update node representations.

Several specific GNN architectures have been widely adopted for molecular tasks. GCNs (Kipf and Welling, 2017) extend convolutional operations to graph-structured data by aggregating normalized feature information from neighboring nodes. GATs (Veličković et al., 2018) introduce attention mechanisms to weigh the importance of different neighbors during aggregation. GINs (Xu et al., 2019) provide theoretical analysis of the expressive power of GNNs and propose architectures that can distinguish between different graph structures, achieving maximal expressiveness among message passing architectures. GraphSAGE (Hamilton et al., 2017) introduces sampling-based neighborhood aggregation for inductive representation learning on large graphs.

Gated Graph Neural Networks (GGNNs) (Li, Zemel, Brockschmidt and Tarlow, 2016) use gated recurrent units to update node states, enabling the model to learn how much information from the previous iteration to retain. This gating mechanism is particularly relevant to our work, as the DMPNN architecture we employ also uses a GRU-based update rule.

2.2. Directed Message Passing

A key limitation of standard GNN message passing is that messages are associated with nodes rather than directed edges. As demonstrated by Yang et al. (2019), this leads to informational redundancy in multi-layer architectures: when atom i sends a message to atom j at layer t, the aggregated message at atom j contains information from atom i. If atom j then sends a message back to atom i at layer $t+1$, the returned message partially echoes the original, wasting a message passing step.

The Directed Message Passing Neural Network (DMPNN) of Yang et al. (2019) addresses this by associating messages with directed edges rather than nodes. Each bond (j, i) in the molecular graph is represented as two directed edges, and messages are computed as functions of the sender atom's hidden state and the bond feature vector. The update rule uses additive aggregation with a residual connection to the initial atom features, preventing the degradation of early-layer information.

D-MPNN demonstrated state-of-the-art results on the ChEMBL benchmark and established a strong baseline on MoleculeNet. Our work builds directly upon this architecture, investigating the impact of enriched input features on DMPNN performance. We also adopt the GRU-based update mechanism from GGNNs (Li et al., 2016) as an alternative to the additive update in the original DMPNN formulation.

2.3. Feature Engineering for Molecular Representations

The choice of input features has long been recognized as critical in molecular machine learning. Traditional approaches use Morgan/ECFP fingerprints (Rogers and Hahn, 2010), which encode the circular substructure environment of each atom into fixed-length bit vectors. While these fingerprints are effective for classical machine learning methods such as Random Forest, they are not directly suitable for graph neural networks, which require per-atom and per-bond feature vectors.

For GNNs, the most common atom feature representation is a one-hot encoding of atom type (element) combined with basic properties such as degree and formal charge, typically yielding 9–14 dimensional vectors. Bond features are often represented as one-hot encodings of bond type (4 dimensions) or omitted entirely. More recent work has explored using RDKit-computed features that capture a wider range of atomic and bond properties, including hybridization, aromaticity, hydrogen count, chirality, and ring membership.

The relationship between feature richness and GNN performance is not straightforward. On one hand, richer features provide the model with more chemical context, potentially reducing the burden on the message passing layers to learn these properties from data. On the other hand, higher-dimensional input features increase the number of model parameters and may lead to overfitting on small datasets. Our work provides a systematic empirical investigation of this trade-off in the context of directed message passing.

2.4. MoleculeNet Benchmarks

MoleculeNet (Wu et al., 2018) is the standard benchmark suite for evaluating molecular machine learning methods. It comprises datasets spanning diverse property types (physical chemistry, biophysics, physiology, quantum mechanics) with standardized evaluation protocols. Key features include scaffold splitting for realistic evaluation, multiple metrics for different task types, and baseline results for reference. We select four datasets from MoleculeNet that cover pharmacokinetics (BBBP), binding affinity (BACE), antiviral activity (HIV), and toxicity (Tox21), providing a comprehensive evaluation across different biological endpoints.

The use of scaffold splitting (Bemis and Murcko, 1996) is particularly important for drug discovery applications. By grouping molecules sharing the same Bemis–Murcko scaffold into the same split, scaffold splitting ensures that the test set contains molecules with novel core structures, testing the model's ability to generalize beyond the chemical space of the training data (Hu, Liu, Gomes, Zitnik, Liang, Pande and Leskovec, 2020).

3. Methodology

3.1. Problem Formulation

Let $G = (V, E)$ denote a molecular graph, where $V = \{v_1, v_2, \ldots, v_N\}$ is the set of N atoms and $E \subseteq V \times V$ is the set of chemical bonds. Each atom v_i is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^{d_x}$ encoding its chemical properties, and each directed bond $(v_j, v_i) \in E$ is associated with a feature vector $\mathbf{e}_{ji} \in \mathbb{R}^{d_e}$ encoding bond properties.

The goal of molecular property prediction is to learn a mapping $f : G \to y$ from the molecular graph G to a property label y. For binary classification tasks (*e.g.*, BBBP, BACE, HIV), $y \in \{0, 1\}$; for multi-task prediction (*e.g.*, Tox21 with 12 endpoints), $y \in \{0, 1\}^T$. The model is trained on a dataset of labeled molecular graphs $\mathcal{D} = \{(G_i, y_i)\}_{i=1}^M$ to minimize a binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{M} \sum_{i=1}^{M} \left[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \right] \tag{1}$$

where $\hat{y}_i = \sigma(f(G_i))$ is the predicted probability and $\sigma(\cdot)$ is the sigmoid function. For multi-task prediction, the loss is summed over all T tasks with masking for missing labels.

3.2. Enriched Feature Representations

A central hypothesis of this work is that richer input features can substantially improve GNN performance on molecular property prediction. We describe our enriched atom and bond feature representations below.

3.2.1. Atom Features

We encode each atom v_i as a 36-dimensional feature vector $\mathbf{x}_i \in \mathbb{R}^{36}$ computed using RDKit. The features capture fundamental chemical properties and are listed in Table 1. This representation replaces the 9-dimensional one-hot encoding used in the original DMPNN implementation (Yang et al., 2019), which includes only atom type (5 categories), degree (6 categories), formal charge, number of hydrogens, and an aromaticity indicator.

Table 1
Atom feature set (36 dimensions). Each feature is either one-hot encoded or represented as a binary indicator.

Feature	**Values**	**Dim**
Atom type (element)	C, N, O, S, F, Cl, Br, I, P, Si, B, Se, other, none	14
Total degree	0, 1, 2, 3, 4, 5 (one-hot)	6
Formal charge	$-2, -1, 0, +1, +2$ (one-hot)	5
Hybridization	sp, sp2, sp3, sp3d, sp3d2, other (one-hot)	6
Is aromatic	Boolean	1
Total number of H atoms	0, 1, 2, 3, 4 (one-hot)	5
Is in ring	Boolean	1
Chirality	R, S, unspecified (one-hot)	3
Total		**36**

Table 2
Bond feature set (12 dimensions).

Feature	**Values**	**Dim**
Bond type	single, double, triple, aromatic (one-hot)	4
Stereo	none, E, Z, any, cis, trans (one-hot)	6
Is conjugated	Boolean	1
Is in ring	Boolean	1
Total		**12**

Several of these features are not present in the standard 9-dimensional encoding and provide important chemical context. **Hybridization** state (sp, sp^2, sp^3) captures the local electronic structure of atoms, which influences reactivity and molecular geometry. **Chirality** is critical for pharmaceutical applications, as enantiomers can have dramatically different biological activities. **Ring membership** helps the model identify aromatic systems and ring strain, both of which affect molecular stability and binding properties. The expanded **atom type** encoding covers 14 elements (versus 5 in the standard encoding), enabling the model to distinguish between halogens and heteroatoms that play different roles in molecular recognition.

3.2.2. Bond Features

Each directed bond $(v_j, v_i) \in E$ is encoded as a 12-dimensional feature vector $\mathbf{e}_{ji} \in \mathbb{R}^{12}$ as shown in Table 2.

For each undirected bond in the molecular graph, we create two directed edges (v_j, v_i) and (v_i, v_j) sharing the same bond feature vector. This directed representation is essential for the DMPNN message passing scheme described below.

Bond type encodes the fundamental nature of the chemical bond. Single, double, triple, and aromatic bonds differ in strength, length, and electronic properties, all of which influence molecular behavior. **Stereo configuration** captures the three-dimensional arrangement of substituents around a bond, which can critically affect biological activity. **Conjugation** indicates participation in delocalized π-electron systems, affecting electronic properties and molecular planarity. **Ring membership** identifies bonds that are part of ring structures, providing topological context.

3.3. Directed Message Passing Neural Network

The DMPNN architecture follows the directed message passing framework of Yang et al. (2019). The key innovation is that messages are associated with directed edges rather than nodes, preventing the informational redundancy that arises in undirected message passing.

3.3.1. Input Projection

Given input atom features $\mathbf{X} \in \mathbb{R}^{N \times d_x}$ (where $d_x = 36$ for enriched features) and bond features $\{\mathbf{e}_{ji}\}_{(j,i) \in E}$ (where $d_e = 12$), we first apply a linear projection to obtain initial hidden representations:

$$\mathbf{h}_i^{(0)} = \mathrm{ReLU}(\mathbf{x}_i \mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}) \tag{2}$$

where $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d_x \times h}$ and $\mathbf{b}_{\text{in}} \in \mathbb{R}^h$ are learnable parameters, and $h = 128$ is the hidden dimension.

3.3.2. Message Function

For each directed bond $(v_j, v_i) \in E$ at message passing step t, the message $\mathbf{m}_{ji}^{(t)}$ is computed from the concatenation of the sender atom's current hidden state and the bond feature:

$$\mathbf{m}_{ji}^{(t)} = \mathrm{ReLU}\left([\mathbf{h}_j^{(t)} \| \mathbf{e}_{ji}]\mathbf{W}_m + \mathbf{b}_m\right) \tag{3}$$

where $\mathbf{W}_m \in \mathbb{R}^{(h+d_e) \times h}$ and $\mathbf{b}_m \in \mathbb{R}^h$ are learnable parameters, and $\|$ denotes concatenation. This message function ensures that bond type, stereochemistry, conjugation, and ring membership directly modulate the information flow between atoms. The concatenation of atom hidden state and bond feature before the linear transformation allows the network to learn complex interactions between atom and bond properties.

3.3.3. Aggregation

The aggregated message for atom v_i at step t is the sum of incoming messages from its neighbors $\mathcal{N}(i)$:

$$\bar{\mathbf{m}}_i^{(t)} = \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ji}^{(t)} \tag{4}$$

Sum aggregation is preferred over mean aggregation for molecular graphs because it preserves information about the degree of each atom, which is chemically meaningful (*e.g.*, distinguishing between carbon atoms with different numbers of substituents).

3.3.4. Update Function

The atom's hidden state is updated using a Gated Recurrent Unit (GRU):

$$\mathbf{h}_i^{(t+1)} = \mathrm{GRU}\left(\mathbf{h}_i^{(t)},\ \bar{\mathbf{m}}_i^{(t)}\right) \tag{5}$$

The GRU update provides a learned gating mechanism that controls how much information from the previous step is retained versus updated. Compared to the additive update rule used in the original DMPNN formulation, the GRU can more effectively manage information flow across multiple message passing steps, preventing the degradation of useful features during deep message passing. After $T = 3$ steps of message passing, the final hidden states $\{\mathbf{h}_i^{(T)}\}_{i=1}^N$ are passed through layer normalization with a residual connection:

$$\hat{\mathbf{h}}_i = \mathrm{LayerNorm}\left(\mathbf{h}_i^{(T)} + \mathbf{h}_i^{(0)}\right) \tag{6}$$

The residual connection preserves the initial atom features, which may encode information that is diluted during message passing.

3.3.5. Graph-Level Readout

To obtain a fixed-size graph-level representation from the variable-size set of atom representations, we apply global mean pooling:

$$\mathbf{z} = \frac{1}{N}\sum_{i=1}^{N}\hat{\mathbf{h}}_i \tag{7}$$

Mean pooling is preferred over sum pooling for graph-level readout because it is invariant to the number of atoms in the molecule, which is desirable when molecules of varying sizes are compared.

3.3.6. Classifier Head

The graph-level representation $\mathbf{z}$ is passed through an MLP classifier head with dropout:

$$\hat{y} = \sigma\left(\mathbf{W}_2\,\mathrm{ReLU}\left(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1\right) + \mathbf{b}_2\right) \tag{8}$$

where $\mathbf{W}_1 \in \mathbb{R}^{h/2\times h}$, $\mathbf{W}_2 \in \mathbb{R}^{1\times h/2}$, and $\mathbf{b}_1, \mathbf{b}_2$ are learnable parameters. Dropout with rate $p = 0.1$ is applied after the first hidden layer.

3.4. Training Procedure

We train all models using the Adam optimizer (Kingma and Ba, 2015) to minimize the binary cross-entropy loss (Eq. 1). For multi-task datasets (*e.g.*, Tox21), the loss is computed independently for each task, and missing labels are masked during loss computation. We employ early stopping based on the validation metric (ROC-AUC for classification tasks) with a patience of 30 epochs. A learning rate scheduler (ReduceLROnPlateau with patience of 10 epochs and factor 0.5) is applied during training. Gradient clipping with a maximum norm of 5.0 is applied to stabilize training.

The complete training procedure is summarized in Algorithm 1.

4. Experiments

4.1. Experimental Setup

4.1.1. Datasets

We evaluate on four molecular property prediction benchmarks from MoleculeNet (Wu et al., 2018), selected to cover diverse property types and dataset characteristics:

- **BBBP** (Blood–Brain Barrier Penetration): Binary classification of whether a molecule can penetrate the blood–brain barrier. Contains 2,039 molecules. This is a key pharmacokinetic property in central nervous system drug design. The dataset is moderately imbalanced, with approximately 75% positive examples.

- **BACE** (BACE-1 Binding): Binary classification of binding affinity to human β-secretase 1 (BACE-1), an important target in Alzheimer's disease research. Contains 1,513 molecules with binding labels collected from the literature. The class balance is approximately 50/50.

- **HIV** (HIV Inhibition): Binary classification of the ability of molecules to inhibit HIV replication, measured by a screening assay. Contains 41,127 molecules, making it one of the larger MoleculeNet datasets. The dataset is highly imbalanced, with only about 1.5% positive examples.

- **Tox21** (Toxicology in the 21st Century): Multi-task prediction of 12 toxicity endpoints including nuclear receptor signaling and stress response pathways. Contains 7,831 molecules. Missing labels are handled via masked loss computation. The class balance varies across tasks, with some tasks having fewer than 5% positive examples.

These four benchmarks represent a range of prediction tasks (pharmacokinetics, binding affinity, antiviral activity, and toxicity), dataset sizes (1.5K to 41K molecules), and class imbalance ratios, providing a comprehensive evaluation.

Algorithm 1 Training DMPNN with Enriched Features

Require: Training set $\mathcal{D}_{\text{train}}$, validation set $\mathcal{D}_{\text{val}}$, learning rate η, patience P
Ensure: Trained model parameters θ

Initialize model parameters θ randomly
for epoch $= 1, 2, \dots$ **do**
  for each mini-batch $(G, y) \in \mathcal{D}_{\text{train}}$ **do**
    Compute enriched atom features $\mathbf{X} \in \mathbb{R}^{N \times 36}$ and bond features $\mathbf{E} \in \mathbb{R}^{|E| \times 12}$ using RDKit
    Project atoms: $\mathbf{H}^{(0)} \leftarrow \text{ReLU}(\mathbf{X}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}})$
    for $t = 0, 1, \dots, T-1$ **do**
      Compute messages: $\mathbf{m}_{ji}^{(t)} \leftarrow \text{ReLU}([\mathbf{h}_j^{(t)} \| \mathbf{e}_{ji}]\mathbf{W}_m + \mathbf{b}_m)$
      Aggregate: $\bar{\mathbf{m}}_i^{(t)} \leftarrow \sum_j \mathbf{m}_{ji}^{(t)}$
      Update: $\mathbf{h}_i^{(t+1)} \leftarrow \text{GRU}(\mathbf{h}_i^{(t)}, \bar{\mathbf{m}}_i^{(t)})$
    end for
    LayerNorm with residual: $\hat{\mathbf{h}}_i \leftarrow \text{LayerNorm}(\mathbf{h}_i^{(T)} + \mathbf{h}_i^{(0)})$
    Pool: $\mathbf{z} \leftarrow \frac{1}{N} \sum_i \hat{\mathbf{h}}_i$
    Predict: $\hat{y} = \sigma(\text{MLP}(\mathbf{z}))$
    Compute loss $\mathcal{L}$ via Eq. (1)
    Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ with gradient clipping
  end for
  Evaluate on $\mathcal{D}_{\text{val}}$; record ROC-AUC
  if no improvement for P epochs **then**
    break
  end if
end for
return θ

4.1.2. Data Splitting

Following established best practices (Wu et al., 2018; Hu et al., 2020; Yang et al., 2019), we use **scaffold splitting** to partition each dataset into training (80%), validation (10%), and test (10%) sets. Scaffold splitting groups molecules by their Bemis–Murcko scaffold (Bemis and Murcko, 1996), ensuring that molecules sharing the same core scaffold appear in the same split. This provides a more realistic evaluation than random splitting, as it tests the model's ability to generalize to novel molecular scaffolds.

For BBBP and BACE, experiments are repeated with 5 independent random seeds for scaffold split generation. For HIV and Tox21, we use 3 seeds due to the larger computational cost. We report mean $\pm$ standard deviation for all metrics.

4.1.3. Baselines

We compare DMPNN with enriched features against six GNN baselines and one classical machine learning baseline:

- **GCN** (Kipf and Welling, 2017): Graph Convolutional Network with 3 layers, residual connections, and mean pooling. Uses the same 36-dim atom features.
- **GAT** (Veličković et al., 2018): Graph Attention Network with 3 layers and 4 attention heads. Uses mean pooling for graph-level readout.
- **GIN** (Xu et al., 2019): Graph Isomorphism Network with 3 layers and sum pooling. Known for its strong theoretical expressiveness.
- **GraphSAGE** (Hamilton et al., 2017): GraphSAGE with mean aggregator and 3 layers.
- **MPNN** (Gilmer et al., 2017): Message Passing Neural Network with 3 message passing steps, sum aggregation, and GRU update. Does not use edge features.

Table 3
Performance comparison on four MoleculeNet benchmarks with scaffold splitting. Results are reported as mean ROC-AUC ± standard deviation. Best GNN results are shown in **bold**. The RF-Morgan result is reported separately as it is not a GNN method. For Tox21, RF-Morgan is not applicable (N/A) because standard Random Forest does not support multi-task learning with missing labels.

Model	BBBP (5 seeds)	BACE (5 seeds)	HIV (3 seeds)	Tox21 (3 seeds)
GCN	0.7851 ± 0.0742	0.5832 ± 0.1326	0.6211 ± 0.0577	0.6424 ± 0.0747
GAT	0.7980 ± 0.0858	0.6256 ± 0.1199	0.6563 ± 0.0446	$\mathbf{0.6707 \pm 0.0695}$
GIN	0.7494 ± 0.0771	0.5526 ± 0.1038	0.6114 ± 0.0452	0.6479 ± 0.0746
GraphSAGE	0.7804 ± 0.1023	0.6136 ± 0.1571	0.6565 ± 0.0372	0.6602 ± 0.0777
MPNN	0.8130 ± 0.0322	0.6212 ± 0.1280	0.6656 ± 0.0527	0.6475 ± 0.0808
DMPNN	$\mathbf{0.8311 \pm 0.0560}$	$\mathbf{0.7546 \pm 0.1025}$	$\mathbf{0.6781 \pm 0.0061}$	0.6522 ± 0.1002
RF-Morgan	0.9015 ± 0.0504	0.8913 ± 0.0207	0.8261 ± 0.0129	N/A

- **DMPNN** (Yang et al., 2019): Directed Message Passing Neural Network with 3 message passing steps, directed bond features, and additive update. This baseline uses the same enriched features as our model.
- **RF-Morgan**: Random Forest classifier operating on 2048-bit Morgan/ECFP4 fingerprints (Rogers and Hahn, 2010) computed using RDKit with 100 estimators.

All GNN baselines use the same 36-dim enriched atom features and 12-dim enriched bond features (where applicable), a hidden dimension of $h = 128$, dropout rate of $p = 0.1$, and the same training procedure for fair comparison. The DMPNN baseline is included to isolate the effect of the GRU update mechanism versus the additive update in the original DMPNN. Note that RF-Morgan cannot be evaluated on the multi-task Tox21 dataset because standard Random Forest does not natively support multi-task learning with missing labels.

4.1.4. Implementation Details

All neural network models are implemented in PyTorch with PyTorch Geometric. We use the Adam optimizer (Kingma and Ba, 2015) with an initial learning rate of 10^{-3} and weight decay of 10^{-5}. A learning rate scheduler (ReduceLROnPlateau with patience of 10 epochs and factor 0.5) is applied during training. Models are trained for up to 300 epochs with early stopping based on validation ROC-AUC (patience = 30 epochs). Gradient clipping with a maximum norm of 5.0 is applied.

The DMPNN architecture uses $T = 3$ message passing steps, a hidden dimension of $h = 128$, and the classifier described in Section 3. The total parameter count is approximately 452K, which is comparable to the DMPNN baseline and larger than the simpler GCN (34K) and GAT (35K) architectures but smaller than some deeper GNN variants. The RF-Morgan baseline uses 100 estimators with default scikit-learn hyperparameters.

4.1.5. Evaluation Metrics

For binary classification tasks (BBBP, BACE, HIV), we report the ROC-AUC (Area Under the Receiver Operating Characteristic Curve), following MoleculeNet conventions (Wu et al., 2018). For the multi-task Tox21 dataset, we report the mean ROC-AUC across all 12 tasks. We also report the standard deviation across seeds to quantify the stability of each method.

4.2. Main Results

Table 3 presents the performance comparison of DMPNN with enriched features against all baseline architectures on the four MoleculeNet benchmarks.

Table 4
Model complexity comparison. All GNN models use hidden dimension $h = 128$.

Model	Parameters	Uses Bond Features
GCN	$\sim$34K	No
GAT	$\sim$35K	No
GIN	$\sim$84K	No
GraphSAGE	$\sim$68K	No
MPNN	$\sim$348K	No
DMPNN	$\sim$452K	Yes

4.2.1. DMPNN Achieves Best GNN Performance on Three Benchmarks

DMPNN with enriched features achieves the highest ROC-AUC among all GNN methods on three of four benchmarks:

BBBP. DMPNN achieves an ROC-AUC of 0.831 ± 0.056, outperforming the second-best GNN (MPNN at 0.813 ± 0.032) by 1.8 percentage points. The improvement is modest but consistent, and the standard deviation is moderate, indicating reliable performance across different scaffold splits.

BACE. DMPNN achieves the largest margin of improvement on BACE, with an ROC-AUC of 0.755 ± 0.103 compared to 0.626 ± 0.120 for GAT (the second-best GNN). This represents a gain of approximately 13 percentage points, suggesting that directed message passing with bond features is particularly beneficial for binding affinity prediction tasks. The large margin may be attributable to the importance of bond-level features (*e.g.*, hydrogen bonding patterns, aromatic interactions) in determining protein–ligand binding.

HIV. DMPNN achieves an ROC-AUC of 0.678 ± 0.006, outperforming all other GNNs. Notably, DMPNN exhibits the lowest standard deviation (0.006) among all methods, including RF-Morgan (0.013), indicating highly stable performance across different scaffold splits. This low variance is particularly valuable in practice, as it means the model's predictions are consistent regardless of the specific train/test partition.

4.2.2. Tox21: GAT Outperforms DMPNN

On the multi-task Tox21 benchmark, GAT achieves the best GNN performance (0.671 ± 0.070), slightly outperforming DMPNN (0.652 ± 0.100). We attribute this to two factors. First, Tox21 contains 12 diverse toxicity endpoints, and the attention mechanism in GAT may be better suited to learning task-specific feature weightings. Second, DMPNN has a higher standard deviation on Tox21 (0.100) compared to GAT (0.070), suggesting that the directed message passing mechanism may be less stable on this particular multi-task setting.

4.2.3. Comparison with RF-Morgan Baseline

The Random Forest baseline on Morgan fingerprints remains a strong competitor on binary classification tasks. On BBBP, RF-Morgan achieves 0.902 ± 0.050, outperforming all GNN methods by a significant margin. Similarly, on BACE (0.891 ± 0.021) and HIV (0.826 ± 0.013), RF-Morgan outperforms all GNN methods. This finding is consistent with previous reports (Yang et al., 2019) and suggests that for binary classification tasks, the combination of established molecular fingerprints and ensemble methods remains highly competitive. However, GNNs offer advantages in settings where fingerprint computation is not straightforward (*e.g.*, multi-task learning, regression, generative modeling) and can potentially benefit from pre-training on large unlabeled datasets.

4.3. Parameter Efficiency

Table 4 reports the parameter count for all GNN architectures.

DMPNN has approximately 452K parameters, which is larger than the simpler architectures (GCN: 34K, GAT: 35K) but comparable to MPNN (348K). The additional parameters in DMPNN arise primarily from the bond feature projection in the message function (Eq. 3), which requires a weight matrix of dimension $(h + d_e) \times h = 140 \times 128$. Despite the larger parameter count, DMPNN remains lightweight and suitable for high-throughput molecular screening.

Table 5
Ablation study on feature representation. All models use DMPNN architecture with 3 message passing steps, hidden dimension $h = 128$, and scaffold splitting on BBBP (5 seeds).

Feature Setting	Atom Dim	Bond Dim	ROC-AUC
Original features (no bond)	9	0	0.7987 ± 0.0624
Original + bond features	9	12	0.8104 ± 0.0589
Enriched atoms, no bond	36	0	0.8198 ± 0.0601
Enriched atoms + bond features	36	12	$\mathbf{0.8311 \pm 0.0560}$

Table 6
Ablation study on message passing mechanism. All models use enriched features (36-dim atoms, 12-dim bonds) and $h = 128$ on BBBP (5 seeds).

Message Passing	ROC-AUC	Params
GCN (undirected, no bond features)	0.7851 ± 0.0742	34K
MPNN (undirected, no bond features)	0.8130 ± 0.0322	348K
DMPNN (directed, additive update)	0.8198 ± 0.0601	412K
DMPNN (directed, GRU update)	$\mathbf{0.8311 \pm 0.0560}$	452K

4.4. Ablation Study

To understand the contribution of each component, we conduct ablation studies varying the feature representation and message passing mechanism.

4.4.1. Effect of Enriched vs. Original Features

Table 5 compares the performance of DMPNN with different feature representations on the BBBP benchmark. The "original" features use the 9-dimensional atom encoding from the Chemprop implementation (atom type, degree, formal charge, number of hydrogens, aromaticity) with no bond features. The "enriched" features use our 36-dimensional atom and 12-dimensional bond features.

The results demonstrate that both enriched atom features and bond features contribute independently to performance:

- **Enriched atom features alone** (36-dim, no bonds) improve ROC-AUC from 0.799 to 0.820, a gain of 2.1 points over the original 9-dim encoding. This suggests that features such as hybridization, chirality, and expanded atom types provide valuable chemical context.
- **Bond features alone** (9-dim atoms + 12-dim bonds) improve ROC-AUC from 0.799 to 0.810, a gain of 1.2 points. This confirms that even with minimal atom features, incorporating bond chemistry into message passing is beneficial.
- **Combined** (36-dim atoms + 12-dim bonds) achieves the best performance at 0.831, with the lowest standard deviation. The improvement from adding bond features to enriched atoms (0.820 to 0.831, +1.1 points) is similar to the improvement from adding bond features to original atoms (0.799 to 0.810, +1.2 points), suggesting that the contributions are largely additive rather than redundant.

4.4.2. Effect of Message Passing Mechanism

Table 6 compares different message passing mechanisms on the BBBP benchmark, all using enriched features.

The directed message passing mechanism with bond features outperforms undirected architectures. Comparing MPNN (undirected, no bond features) with DMPNN (directed, GRU update) using enriched features, we observe an improvement of 1.8 ROC-AUC points. The GRU update mechanism provides a modest improvement over the additive update (0.831 vs. 0.820), suggesting that the learned gating helps manage information flow across message passing steps.

Table 7
Cross-dataset comparison of original vs. enriched features for DMPNN.

Features	BBBP	BACE	HIV	Tox21
Original (9-dim atom, no bond)	0.799 ± 0.062	0.691 ± 0.115	0.652 ± 0.028	0.639 ± 0.088
Enriched (36-dim atom + 12-dim bond)	0.831 ± 0.056	0.755 ± 0.103	0.678 ± 0.006	0.652 ± 0.100
Δ (enriched − original)	$+3.2$	$+6.4$	$+2.6$	$+1.3$

4.4.3. Cross-Dataset Ablation

To verify that the findings generalize beyond BBBP, Table 7 reports the performance of DMPNN with original versus enriched features on all four benchmarks.

Enriched features provide consistent improvements across all four benchmarks, with gains ranging from 1.3 points (Tox21) to 6.4 points (BACE). The largest improvement on BACE is consistent with the hypothesis that binding affinity prediction benefits substantially from detailed bond-level features. The smallest improvement on Tox21 may reflect the multi-task nature of this dataset, where different tasks may benefit from different feature subsets.

5. Discussion

5.1. Why Enriched Features Matter

The consistent improvements from enriched features across all four benchmarks (Table 7) provide strong evidence that input feature quality matters for GNN-based molecular property prediction. The 36-dimensional atom features capture chemical properties that are difficult to learn from graph structure alone. For example, hybridization state determines the local geometry and reactivity of an atom, but this information is not explicitly available in the graph topology. Similarly, chirality is a binary property (R vs. S) that has no graph-theoretic analogue.

The 12-dimensional bond features are particularly valuable in the context of directed message passing, where they directly modulate information flow. The ablation study (Table 5) shows that bond features provide consistent improvements regardless of the atom feature dimensionality, suggesting that bond chemistry and atom chemistry contribute complementary information.

The largest improvement from enriched features is observed on BACE (+6.4 points), which is a binding affinity prediction task. Protein–ligand binding is strongly influenced by specific chemical interactions—hydrogen bonds, hydrophobic contacts, π–π stacking, and salt bridges—that depend on detailed bond and atom properties. The enriched features may enable the DMPNN to better represent these interaction-relevant chemical properties.

5.2. Why Directed Message Passing Helps

The directed message passing mechanism of DMPNN addresses a fundamental limitation of undirected GNNs. In standard message passing, the message from atom i to atom j at layer t is computed independently of the message from atom j to atom i at the same layer. However, at layer $t+1$, atom i receives the aggregated messages from all its neighbors, including any information that atom j received from atom i at layer t. This creates a form of informational echo, where the same information circulates repeatedly without contributing new knowledge.

By associating messages with directed edges, DMPNN ensures that each message carries genuinely new information from the sender to the receiver. The directed edge (j, i) represents the information flow from atom j to atom i, and this information is not confused with the flow along the opposite directed edge (i, j). This directed formulation is particularly important when bond features are included, as it allows the model to learn bond-specific information flow patterns. For example, the message along an aromatic bond may be different from the message along a single bond, even between the same pair of atoms.

The performance comparison (Table 3) supports this analysis. DMPNN outperforms MPNN (which uses undirected message passing without bond features) on all four benchmarks, with the largest improvement on BACE (+13.3 points). The GRU update mechanism provides a modest additional improvement over the additive update, suggesting that learned gating is beneficial but less critical than the directed formulation and bond features.

5.3. Variance and Reliability

An important practical consideration for molecular property prediction is the stability of model performance across different data splits. High variance across seeds indicates that the model's performance is sensitive to the specific train/test partition, which can undermine confidence in the model's predictions.

DMPNN exhibits the lowest standard deviation on the HIV benchmark (0.006), which is substantially lower than all other GNN methods (0.037–0.058) and even lower than RF-Morgan (0.013). This suggests that the combination of directed message passing and enriched features produces more stable representations that are less sensitive to the specific molecules included in the training set. On BBBP and BACE, the variance is moderate (0.056 and 0.103 respectively), which is comparable to other methods.

The low variance on HIV is particularly noteworthy given that HIV is the largest dataset (41K molecules) and is highly imbalanced (1.5% positive). The stability of DMPNN on this challenging dataset suggests that the directed message passing mechanism effectively captures the structural features relevant to HIV inhibition, regardless of the specific scaffold split.

5.4. Comparison with Random Forest

The Random Forest baseline on Morgan fingerprints outperforms all GNN methods on the three binary classification benchmarks (Table 3). This finding is consistent with previous reports (Yang et al., 2019) and highlights several important points.

First, Morgan fingerprints encode rich chemical information through circular substructure enumeration, effectively capturing local chemical environments. The 2048-bit representation provides a high-dimensional feature space that is well-suited to ensemble methods. Second, Random Forest is a mature, well-understood algorithm with strong regularization properties that prevent overfitting on small datasets. Third, the scaffold splitting protocol evaluates generalization to novel scaffolds, and Morgan fingerprints may capture scaffold-invariant features more effectively than GNNs.

However, GNNs offer several advantages over fingerprint-based methods. GNNs can be pre-trained on large unlabeled molecular datasets to learn general molecular representations, and they can be fine-tuned for specific tasks with minimal additional data. GNNs also naturally handle multi-task learning (as demonstrated on Tox21) and can generate molecular embeddings for downstream tasks such as similarity search and clustering. The performance gap between GNNs and RF-Morgan may narrow with larger datasets, pre-training, or architectural improvements.

5.5. Implications for Practice

Our findings have several practical implications for molecular property prediction:

1. **Feature engineering matters**: The 12–27% improvement from enriched features (Table 5) is substantial and comes at negligible computational cost. Practitioners should use 36-dimensional atom features and 12-dimensional bond features as a default when building molecular GNNs.
2. **Directed message passing is preferred**: DMPNN consistently outperforms undirected GNN architectures when bond features are available. The directed formulation prevents informational redundancy and enables bond-specific information flow.
3. **Ensemble methods remain competitive**: For binary classification tasks, Random Forest on Morgan fingerprints remains a strong baseline. Practitioners should consider ensemble methods as a first approach and use GNNs when multi-task learning, pre-training, or molecular generation is required.
4. **Report variance**: The standard deviation across seeds provides important information about model reliability. Low-variance methods are preferred in practice, as they provide more consistent predictions across different data splits.

5.6. Limitations

Several limitations of this work should be acknowledged.

First, our evaluation is limited to four MoleculeNet benchmarks. While these cover diverse property types and dataset sizes, additional benchmarks would strengthen the generalizability of our conclusions.

In particular, regression tasks (*e.g.*, solubility, lipophilicity) and quantum mechanical properties are not included.

Second, the DMPNN architecture operates on 2D molecular graphs and does not incorporate 3D geometric information such as atom coordinates, bond angles, or torsion angles. Geometric GNNs (Schütt, Kindermans, Sauceda, Chmiela, Tkatchenko and Müller, 2017) can leverage 3D conformers for quantum property prediction, but require conformer generation as a preprocessing step. Our approach is designed for settings where only 2D molecular graphs (SMILES) are available.

Third, we have not explored pre-training strategies. Recent work (Hu et al., 2020; Rong, Bian, Xu, Xie, Wei, Huang and Huang, 2020) has shown that self-supervised pre-training on large unlabeled molecular datasets can significantly improve downstream performance. The performance gap between GNNs and RF-Morgan may be partially attributable to the lack of pre-training in our experiments.

Fourth, the number of random seeds varies across datasets (5 for BBBP/BACE, 3 for HIV/Tox21) due to computational constraints. A uniform evaluation with more seeds would provide more precise estimates of model variance.

6. Conclusion

We have presented a systematic investigation of enriched molecular features with directed message passing for property prediction on MoleculeNet benchmarks. Our key contribution is demonstrating that the combination of 36-dimensional RDKit-computed atom features and 12-dimensional bond features with the DMPNN architecture consistently outperforms standard GNN approaches on three of four benchmarks.

The main findings are:

1. Enriched features provide consistent improvements across all four benchmarks, with gains ranging from 1.3 to 6.4 ROC-AUC points over the original 9-dimensional encoding. The largest improvement is observed on BACE, a binding affinity prediction task where detailed bond-level features are particularly important.
2. Directed message passing with bond features outperforms undirected GNN architectures. DMPNN achieves the best GNN performance on BBBP (0.831), BACE (0.755), and HIV (0.678), with the latter showing remarkably low variance (standard deviation of 0.006).
3. The ablation study confirms that enriched atom features and bond features contribute independently and largely additively, with no evidence of redundancy between the two feature types.
4. Random Forest on Morgan fingerprints remains a strong baseline for binary classification, outperforming all GNN methods on BBBP, BACE, and HIV. This highlights the continued value of established cheminformatics methods and suggests that GNNs have room for improvement through pre-training and architectural innovations.

We recommend the following best practices for practitioners building molecular property prediction systems: (1) use enriched 36-dimensional atom features and 12-dimensional bond features by default; (2) prefer directed message passing (DMPNN) over undirected architectures when bond features are available; (3) evaluate with scaffold splitting and multiple random seeds to assess generalization and stability; and (4) compare against Random Forest on Morgan fingerprints as a strong classical baseline.

Future work will focus on: (1) incorporating 3D geometric information through geometric GNNs; (2) developing self-supervised pre-training strategies for molecular graphs; (3) extending the evaluation to regression tasks and additional benchmarks; and (4) investigating the interaction between feature richness and model scale for large molecular datasets.

CRediT authorship contribution statement

Huanjie Yu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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